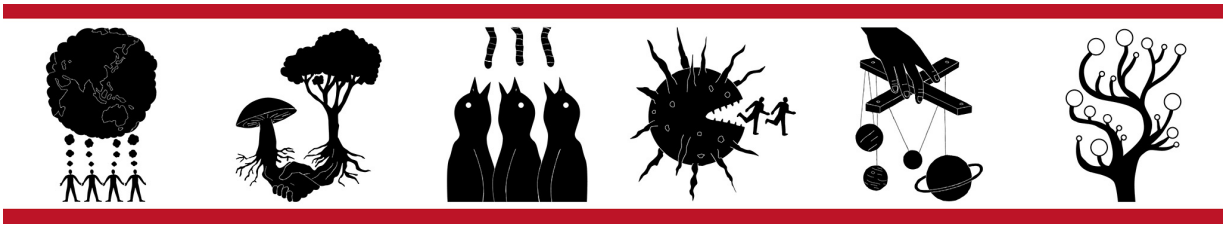
INTRODUCTION



For the earliest humans, a rudimentary knowledge of ecology—how organisms relate to one another—was a matter of life and death. Without having a basic understanding of why animals grazed in a certain place and fruit grew in another, our ancestors would not have survived and evolved.

In the 4th century BCE, Aristotle and his student Theophrastus developed theories of animal metabolism and heat regulation, dissected birds' eggs to learn how they grew, and described an 11-level "ladder of life," the first attempt at classifying organisms.

During the Middle Ages (476–1500), since the Catholic Church discouraged new scientific thought, European understanding of ecology advanced very slowly. By contrast, in the Arab world, many scientific advancements were made during the Islamic Golden Age (8th–13th century). Notably, Abu 'Uthman 'Amr ibn Bahr (known as al-Jahiz) catalogued more than 300 kinds of animals. His notes included not only novel observations, but also the earliest known descriptions of a food chain and of the impact of environmental factors on animals.

By the 16th century, maritime exploration, coupled with great technological advances, such as the invention of the microscope, led to the discovery of amazing life forms and a thirst for knowledge about them. Swedish botanist Carl Linnaeus developed a classification system, *Systema Naturae*, the first scientific attempt to name species and group them according to relatedness. Throughout this time, essentialism—the idea that